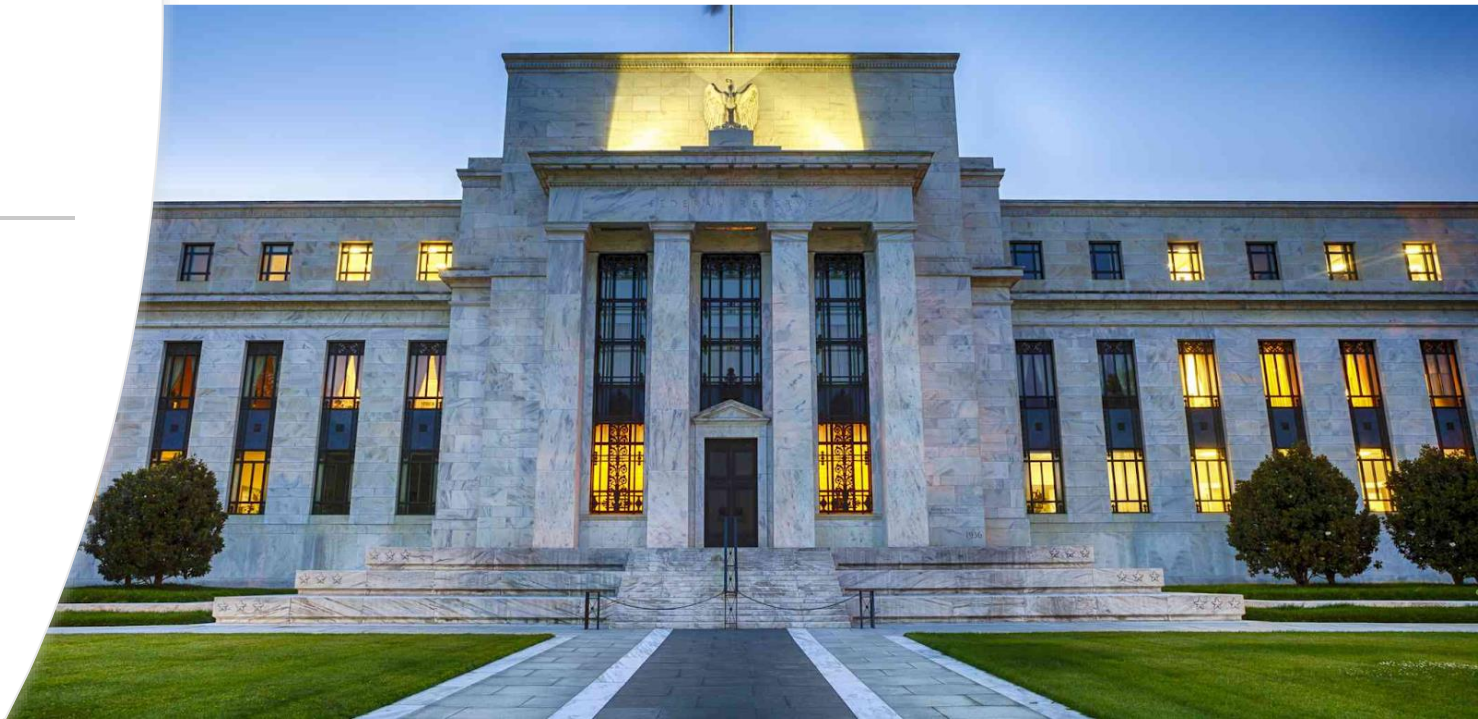
UNC Fed Challenge Club General Meeting

2025-03-04

Jimmy Zheng



Announcements

- No Fed Challenge meeting text week. Spring break!
- In-school Competition:
  - Date: Friday, April 11th
  - Time: TBD,
  - Location: GA309
- Mock Presentation:
  - Team 1 (Captain: Agrima): Tuesday (5:45pm), March 18th, GA309
  - Team 2 (Captain: Sean): Tuesday (5:45pm), March 25th, GA307

Today's Topic: How is the effective federal funds rate set in the federal funds market?

ECONOMIC POLICY REVIEW

How Did the Fed Funds Market Change When Excess Reserves Were Abundant?

Volume 26,, Number 1
March 2020

JEL classification: E4, E5, G2

Authors: John P. McGowan and Ed Nosal

Prior to the 2007-2008 financial crisis, excess reserves in the U.S. banking system were scarce. After the financial crisis and up until early 2018, excess reserves were abundant. In this article, the authors document, and attempt to explain the differences in the performance of the federal funds market under the two different excess reserves frameworks.

Referenced article

The federal funds market

What is the federal funds market?

- An **overnight lending market** where banks borrow and lend **reserves**.
- The **equilibrium interest rate** in this market is called the **effective federal funds rate (EFFR)**.

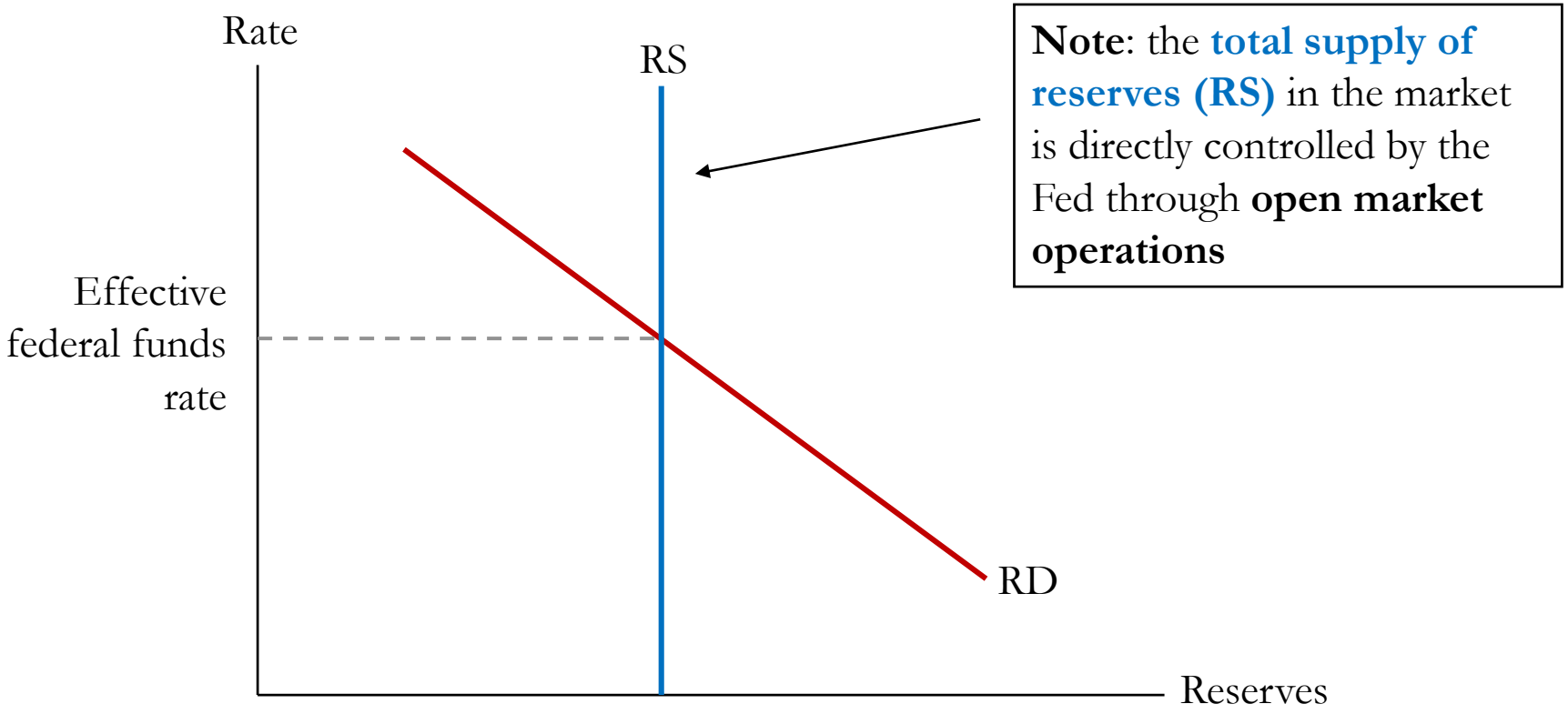
What are bank reserves?

- When you save money at a bank, it becomes your **deposits at that bank**.
- When banks want to save money, they **deposit it at the Fed** (i.e. a central bank OR the bank of banks)
- **Reserves** = Banks' own deposits at the central bank

Why does this market exist?

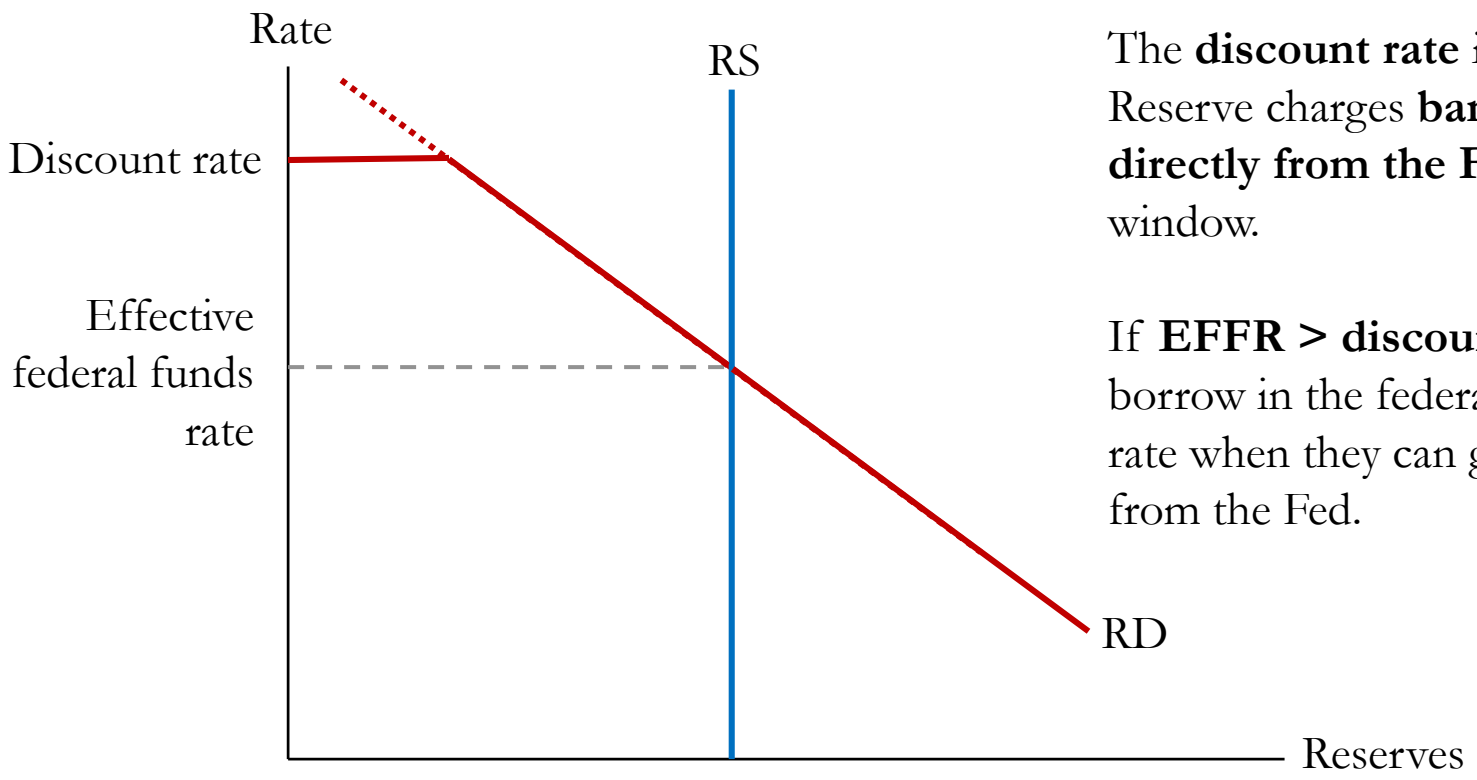
- Banks want to borrow reserves to cover **unexpected liquidity needs** OR to meet the **reserves requirement (RR)** set by the Fed.
- Banks are willing to lend their **excess reserves** (=total reserves balances − required reserves) to earn interest.

The beginning of the story…



Note: the **total supply of reserves (RS)** in the market is directly controlled by the Fed through **open market operations**

Discount rate is an upper bound of EFFR



The **discount rate** is the interest rate the Federal Reserve charges **banks for borrowing reserves directly from the Fed** through the discount window.

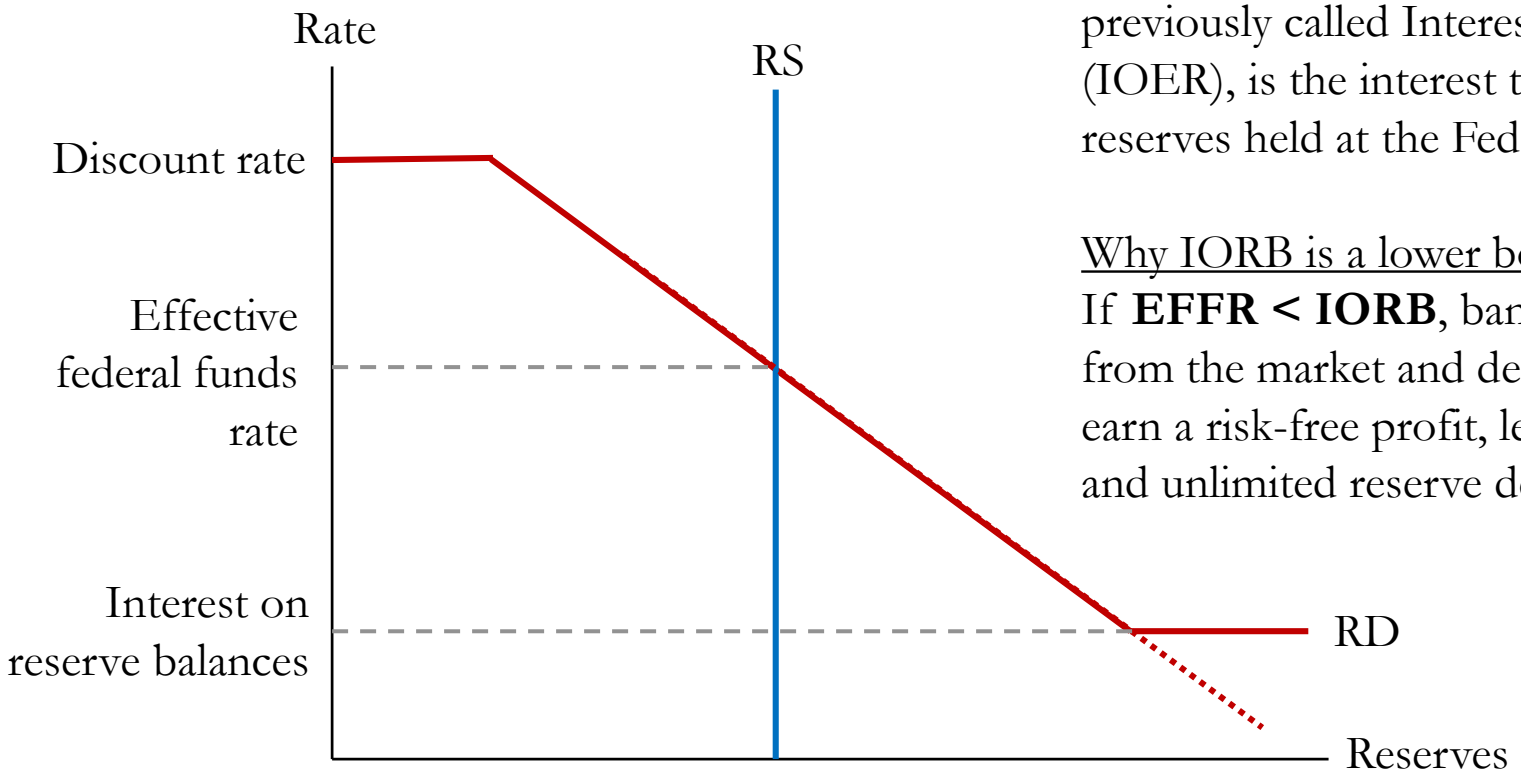
If **EFFR > discount rate**, bank will NOT borrow in the federal funds market at a higher rate when they can get reserves more cheaply from the Fed.

How has the market changed?

Major Shifts Since 2008:

- **Post-2008**: The Fed's multiple rounds of **quantitative easing (QE)** pumped huge excess reserves into banks.
- **October 3, 2008**: The Fed started paying **interest on excess reserves (IOER).**
- **September 23, 2013**: the FOMC introduced the **overnight reverse repurchase (ON RRP)** facility
- **March 26, 2020**: The Fed **eliminated reserve requirements** → No distinction between required vs. excess reserves.

Post-2008: Interest on reserve balances becomes a lower bound

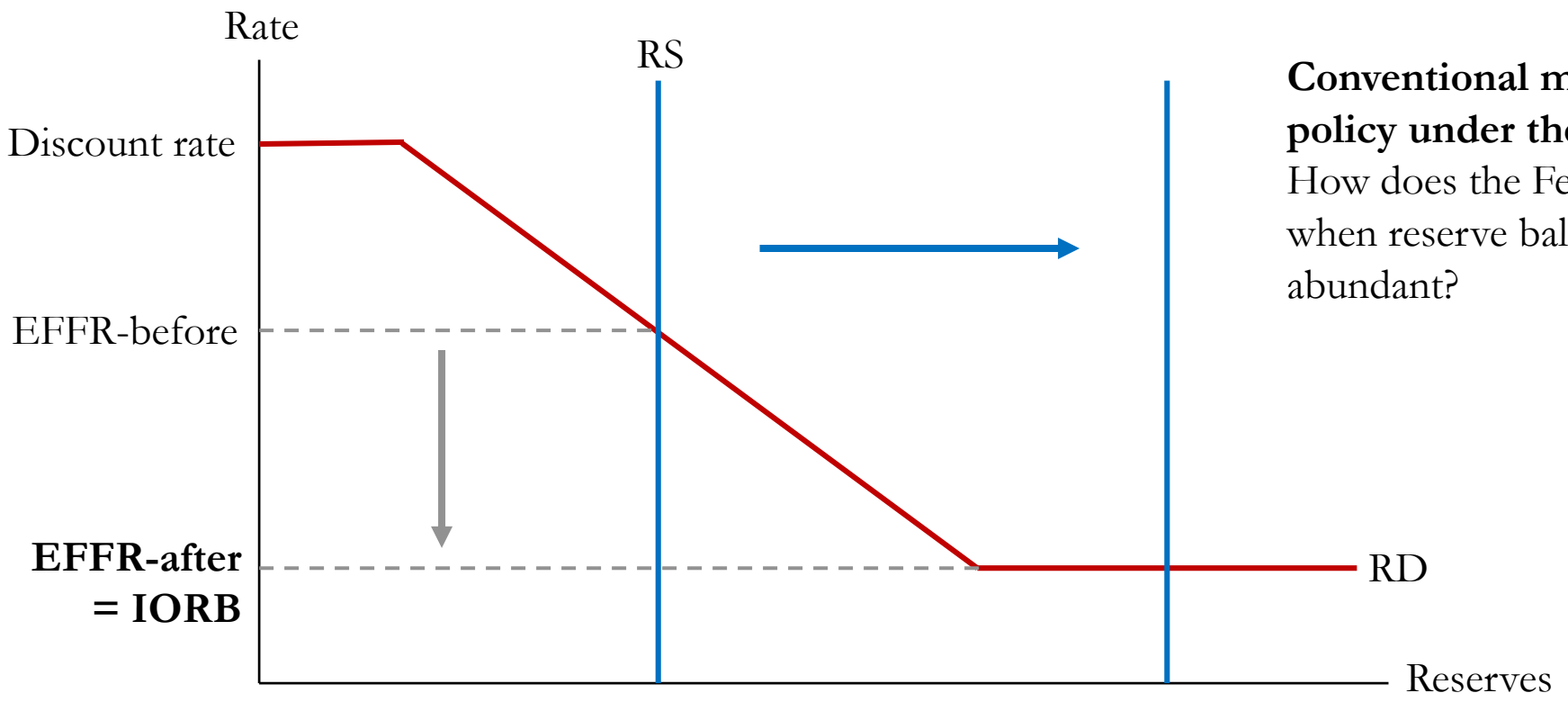


The **Interest on Reserve Balances (IORB)**, previously called Interest on Excess Reserves (IOER), is the interest the Fed pays banks on their reserves held at the Fed.

<u>Why IORB is a lower bound of EFFR?</u>

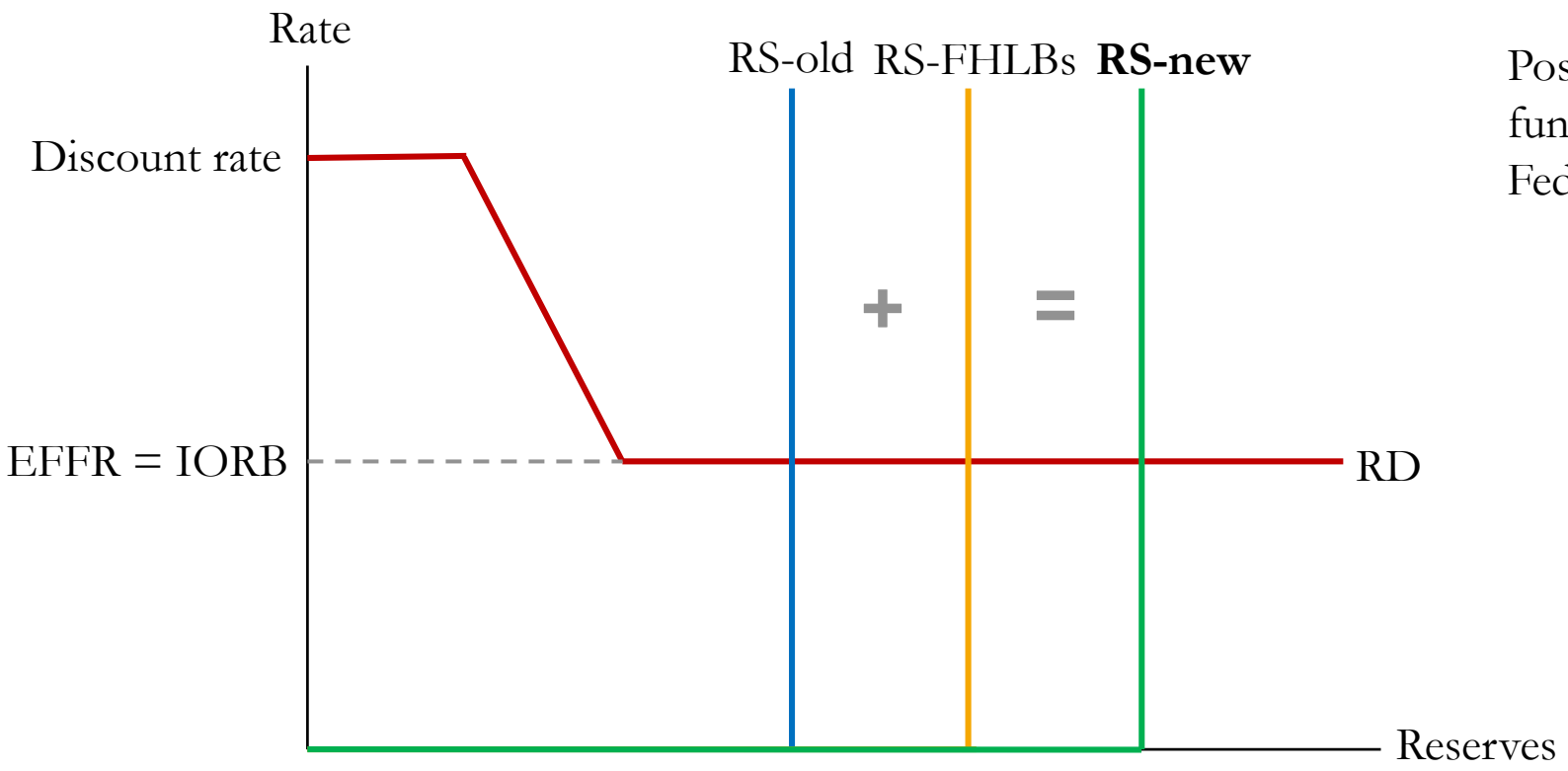
If **EFFR < IORB**, banks could borrow reserves from the market and deposit them at the Fed, and earn a risk-free profit, leading to **IORB-arbitrage** and unlimited reserve demand.

Post-2008 QE increases reserves supply significantly



Conventional monetary policy under the floor system: How does the Fed adjust EFFR when reserve balances are abundant?

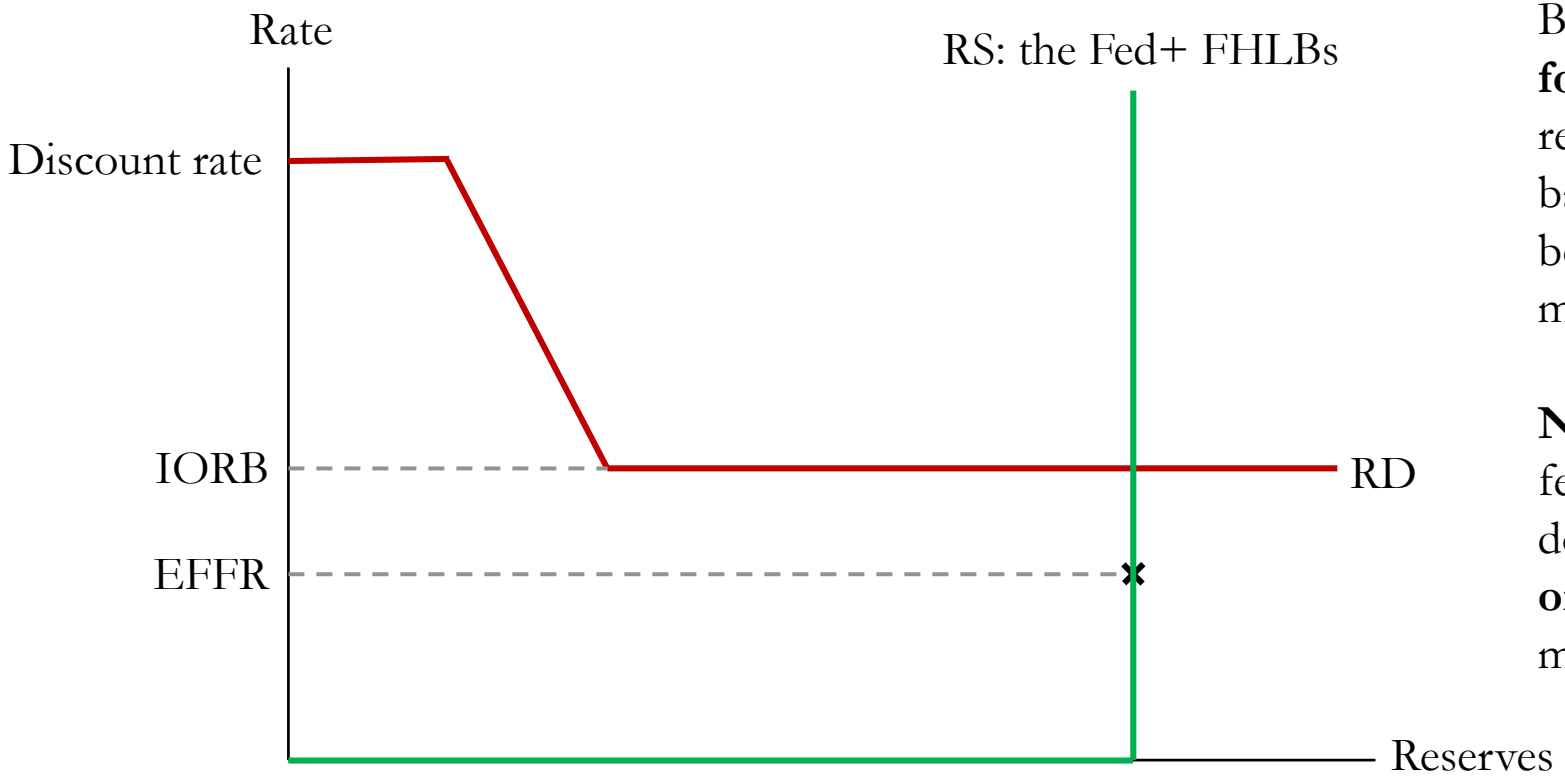
Introduce new lenders of reserves



Post-2008, lending in the federal funds market became dominated by Federal Home Loan Banks (FHLBs):

- FHLBs also hold reserve balances at the Fed
- FHLBs are **ineligible to receive IORB**
- They are willing to supply their total holdings of reserves in the market **whenever EFFR > 0**

IORB-arbitrage pushes EFFR below IORB



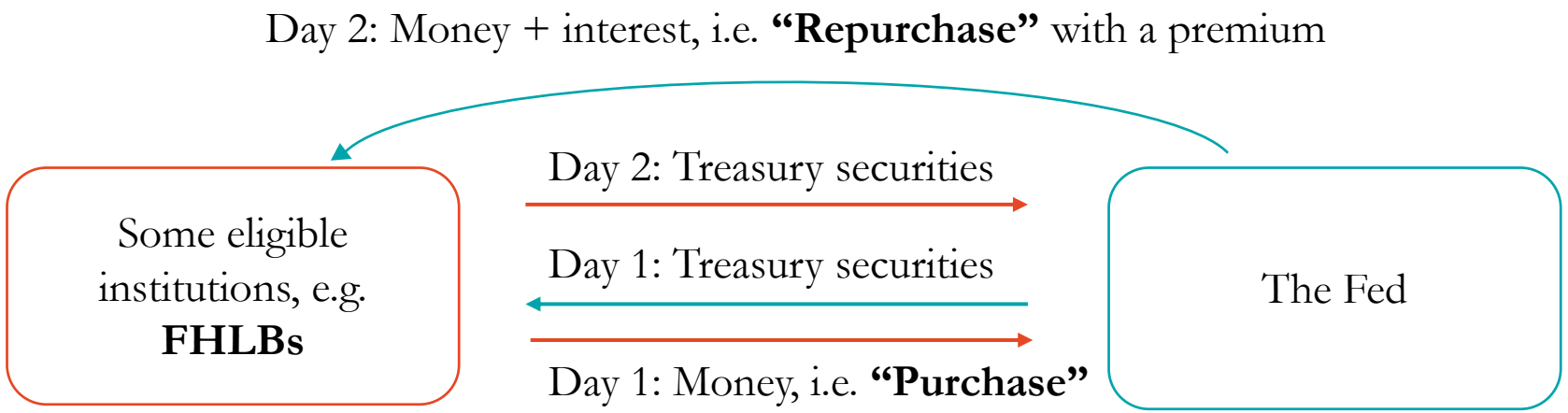
Because **FHLBs are ineligible for IORB**, they are willing to lend reserves at rates below IORB, and banks, driven by this arbitrage, borrow these reserves, which makes **0 < EFFR < IORB**.

Note: Post-2008, borrowing in federal funds market became dominated by **foreign banking organizations (FBOs)**, also motivated by IORB-arbitrage.

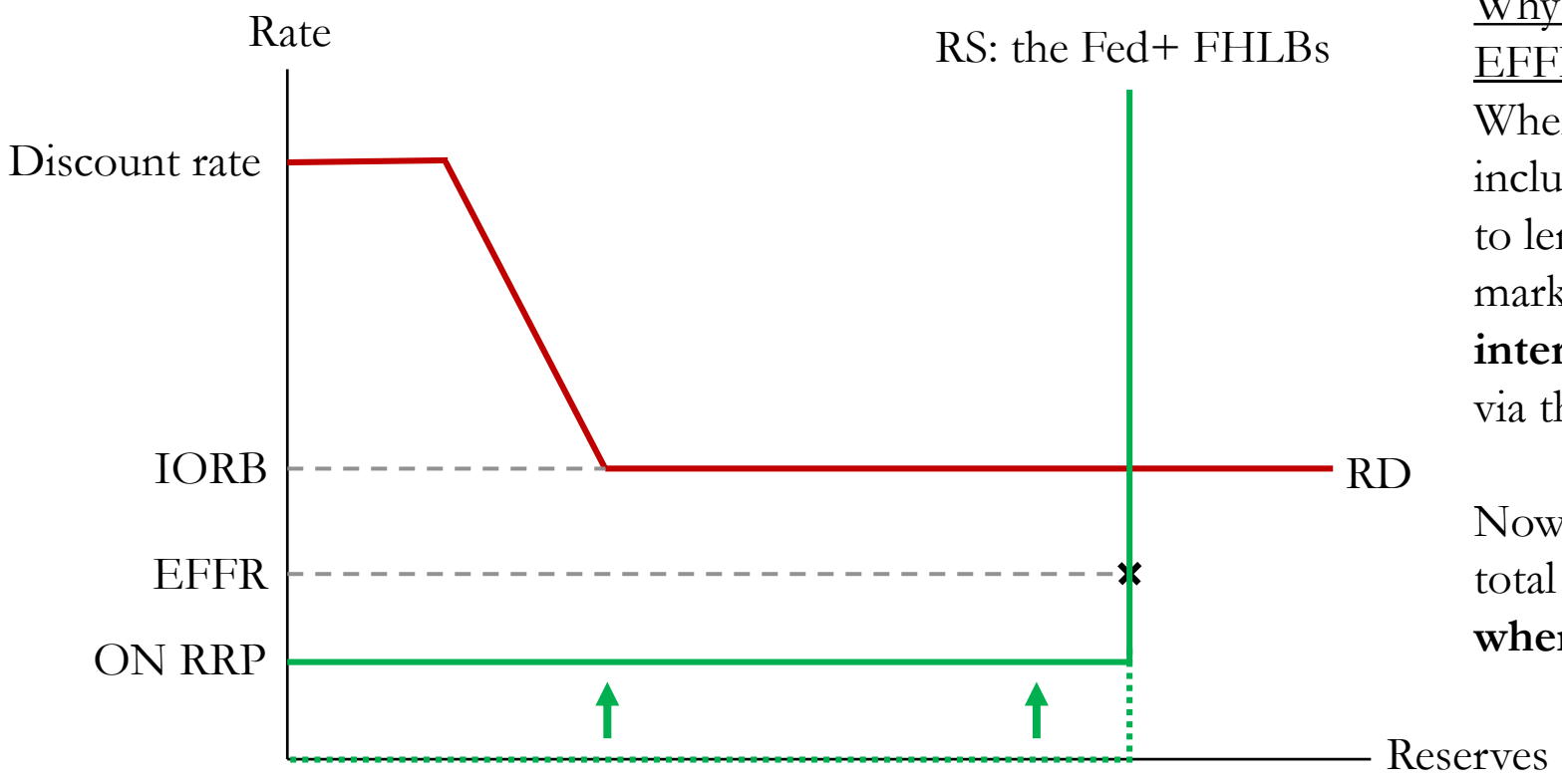
Post-2013: The Fed introduced ON RRP facility

What is the Overnight Reverse Repurchase Agreement (ON RRP)?

- ONRRP facility allows **eligible institutions**, such as money market funds and government-sponsored enterprises, to **lend cash to the Fed overnight** in exchange for Treasury securities as collateral, earning the ON RRP rate.
- **ON RRP = a risk-free overnight lending rate** offered by the Fed to eligible institutions



Post-2013: ON RRP became the new lower bound



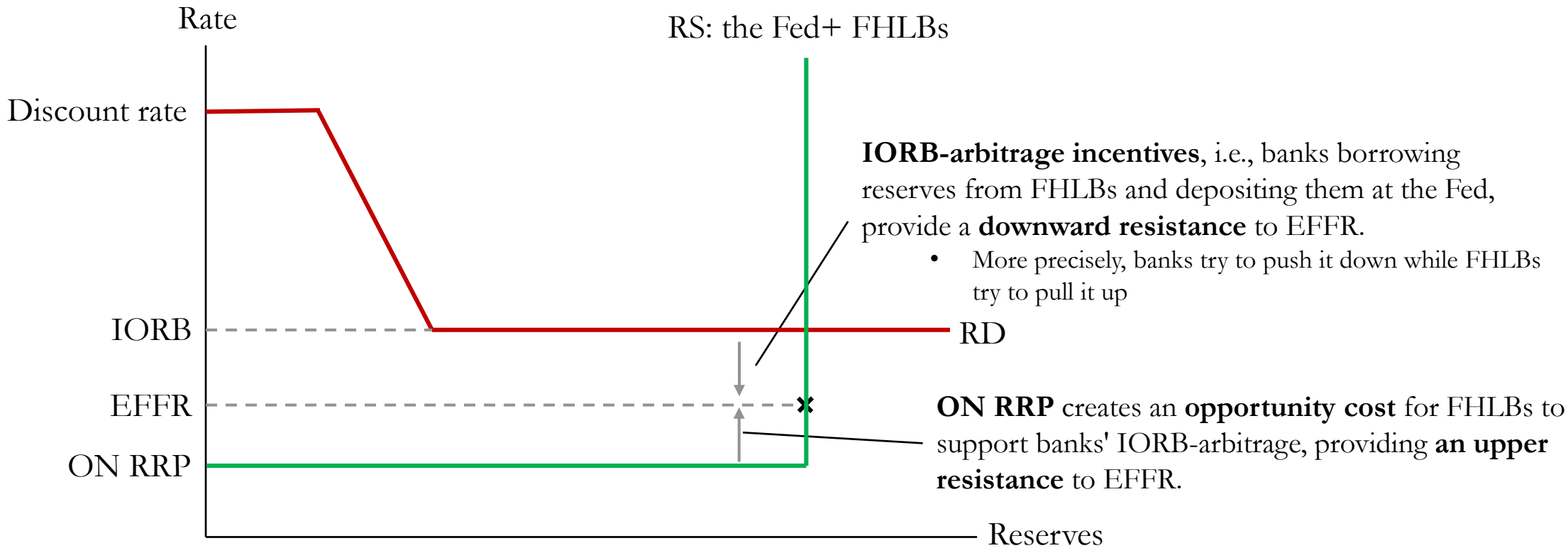
<u>Why ON RRP is an upper bound of EFFR?</u>

When **EFFR < ON RRP**, institutions, including **FHLBs**, will **NOT** be willing to lend reserves in the federal funds market because they can **earn a higher interest rate by lending to the Fed** via the ON RRP facility.

Now, FHLBs are willing to supply their total holdings of reserves in the market **whenever EFFR > ON RRP**.

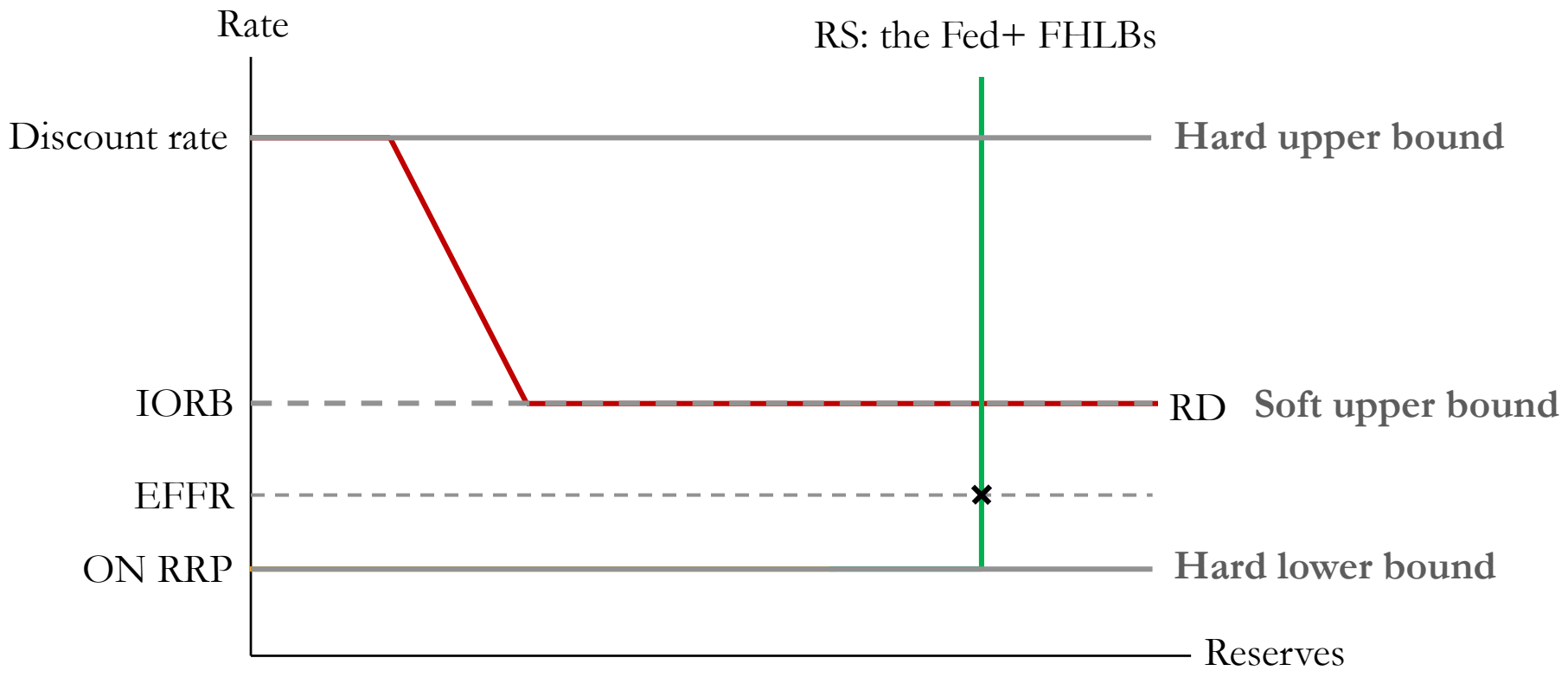
Post-2013: "Arbitrage" determines the effective federal funds rate

Key takeaway 1: Post-2013, **arbitrage incentives**, rather than the usual supply-demand relations, adjust the federal funds market to equilibrium.

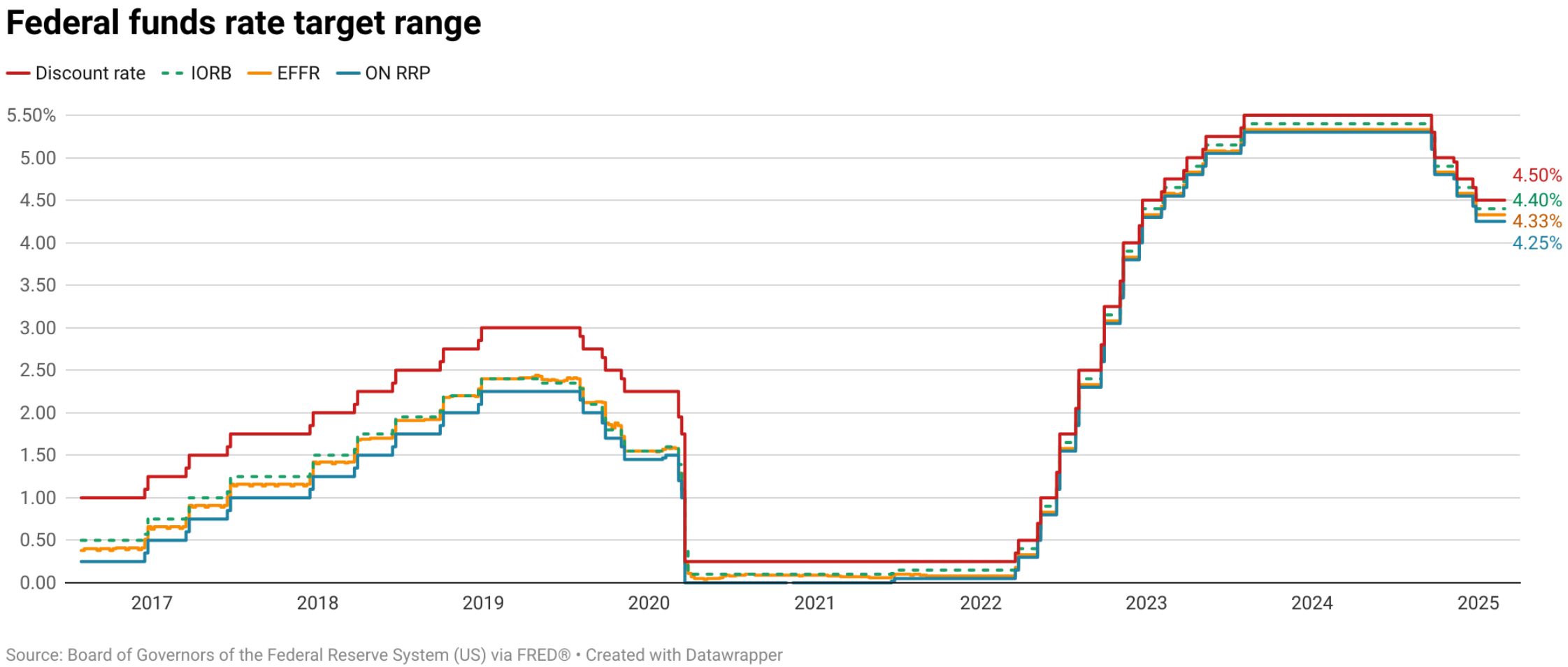


Federal funds rate target range

Key takeaway 2 : Discount rate, IORB, and ON RRP collectively determine the federal funds rate target range.



Data: Federal funds rate target range



Recommended Literature

- McGowan, John P., and Ed Nosal. 2020. *"How Did the Fed Funds Market Change When Excess Reserves Were Abundant?"* Economic Policy Review 26 (1). Federal Reserve Bank of New York.
- Ihrig, Jane E., and Scott A. Wolla. 2020. *"The Fed's New Monetary Policy Tools."* Federal Reserve Bank of St. Louis Page One Economics, August 3.